
Supplement to “Optimal-Dimension U-Calibration by Bayesian Bootstrap”

S1 Dirichlet Laws on Faces

For $c \in [0, \infty)^K \setminus \{0\}$, let $I = \{i : c_i > 0\}$ and $N = \sum_i c_i$. We define D_c on the face

$$\Delta_I = \{p \in \Delta_K : p_i = 0 \text{ for } i \notin I\}$$

by drawing independent $G_i \sim \text{Gamma}(c_i, 1)$ for $i \in I$, setting all other coordinates to zero, and normalizing. Relative to Lebesgue measure on this face, the density is

$$f_c(p) = \frac{\Gamma(N)}{\prod_{i \in I} \Gamma(c_i)} \prod_{i \in I} p_i^{c_i-1}. \quad (\text{S1})$$

If I is a singleton, this notation means the corresponding point mass.

Lemma S1 (Exact density ratio and total variation). *If $c_i > 0$, then*

$$\frac{dD_{c+e_i}}{dD_c}(p) = \frac{Np_i}{c_i}$$

and

$$d_{\text{TV}}(D_c, D_{c+e_i}) = \frac{1}{2} \mathbb{E}_{P \sim D_c} \left| 1 - \frac{NP_i}{c_i} \right| \leq \frac{1}{2} \sqrt{\frac{N - c_i}{c_i(N+1)}}. \quad (\text{S2})$$

If $c_i = 0$, the two distributions are mutually singular unless c is zero, which never arises in this lemma.

Proof. The normalizing constants in (S1) satisfy

$$\frac{\Gamma(N+1)}{\Gamma(N)} \frac{\Gamma(c_i)}{\Gamma(c_i+1)} = \frac{N}{c_i},$$

and the monomial gains a factor p_i . This proves the density ratio. The marginal P_i is $\text{Beta}(c_i, N - c_i)$, with

$$\mathbb{E}P_i = \frac{c_i}{N}, \quad \text{Var}(P_i) = \frac{c_i(N - c_i)}{N^2(N+1)}.$$

The identity in (S2) is the density-ratio formula for total variation. Cauchy–Schwarz and the displayed variance give the upper bound. If $c_i = 0$, $P_i = 0$ under D_c and $P_i > 0$ almost surely under D_{c+e_i} whenever another coordinate is present, so the supports are disjoint up to null sets. $\square$

For later reference, define $H_n^{(1/2)} = \sum_{j=1}^n j^{-1/2}$ and $H_0^{(1/2)} = 0$. Integral comparison gives

$$H_n^{(1/2)} \leq 1 + \int_1^n x^{-1/2} dx = 2\sqrt{n} - 1 \leq 2\sqrt{n}. \quad (\text{S3})$$

S2 Weighted Be the Leader and Exponential Size Bias

We first record the weighted form of be the leader with no regularity assumption on the loss.

Lemma S2 (Weighted be the leader). *Let $w_t \geq 0$, let f_t be arbitrary functions on a common action space, and choose $a_{t+1} \in \arg \min_a \sum_{s=1}^t w_s f_s(a)$. Then*

$$\sum_{t=1}^T w_t f_t(a_{t+1}) \leq \min_a \sum_{t=1}^T w_t f_t(a). \quad (\text{S4})$$

Proof. Let $F_t(a) = \sum_{s=1}^t w_s f_s(a)$. Induction on T gives

$$\begin{aligned} \sum_{t=1}^T w_t f_t(a_{t+1}) &\leq F_{T-1}(a_T) + w_T f_T(a_{T+1}) \\ &\leq F_{T-1}(a_{T+1}) + w_T f_T(a_{T+1}) \\ &= F_T(a_{T+1}) = \min_a F_T(a). \end{aligned}$$

The second line uses the optimality of a_T for F_{T-1} in the direction $F_{T-1}(a_T) \leq F_{T-1}(a_{T+1})$. If some prefix has total weight zero, any action is a leader and the same induction applies. $\square$

For a proper loss and weighted categorical observations, the normalized weighted histogram is a leader. Indeed, if $S = \sum_t w_t > 0$ and $q = S^{-1} \sum_t w_t e_{y_t}$, then

$$\sum_t w_t \ell(p, y_t) = S \sum_i q_i \ell(p, i),$$

which is minimized at $p = q$ by properness.

Lemma S3 (Exponential Palm identity). *Let $W_1, \dots, W_T$ be independent $\text{Exp}(1)$ variables. For every integrable measurable F ,*

$$\mathbb{E}[W_t F(W_1, \dots, W_T)] = \mathbb{E}[F(W_1, \dots, W_t + W'_t, \dots, W_T)], \quad (\text{S5})$$

where W'_t is an independent $\text{Exp}(1)$ variable.

Proof. The density $w e^{-w}$ integrates to one and is the $\text{Gamma}(2, 1)$ density. A sum of two independent unit-rate exponentials has this law. Integrating the other coordinates proves (S5). $\square$

Proposition S4 (Deterministic Dirichlet BTPL inequality). *Fix any outcome path $y_{1:T}$. For every proper loss in $[-1, 1]$,*

$$\sum_{t=1}^T \mathbb{E}_{P \sim D_{c_t}} \ell(P, y_t) - \min_p \sum_{t=1}^T \ell(p, y_t) \leq \sum_{i=1}^K H_{n_i}^{(1/2)}. \quad (\text{S6})$$

Proof. Use one global exponential weight W_t for each observation and define the normalized weighted histogram A_{t+1} through round t . Lemma S2 applies simultaneously to every proper loss. Taking expectations on its right side gives

$$\mathbb{E} \min_p \sum_t W_t \ell(p, y_t) \leq \min_p \sum_t \mathbb{E}[W_t] \ell(p, y_t) = \min_p \sum_t \ell(p, y_t). \quad (\text{S7})$$

Suppose $y_t = i$ and this is its j th occurrence. The classwise sums of the ordinary weights through time t are independent gamma variables with shape vector c_t . Hence $A_{t+1} \sim D_{c_t}$. In the weighted expectation $\mathbb{E}[W_t \ell(A_{t+1}, y_t)]$, Lemma S3 adds one exponential weight to class i . Its normalized histogram has law $D_{c_t + e_i}$. We have proved

$$\sum_t \mathbb{E}_{D_{c_t + e_{y_t}}} \ell(P, y_t) \leq \min_p \sum_t \ell(p, y_t). \quad (\text{S8})$$

For a function in $[-1, 1]$, changing a distribution costs at most twice total variation. Lemma S1 bounds the replacement of $D_{c_t + e_i}$ by D_{c_t} by $1/\sqrt{j}$. Summing by class proves (S6). $\square$

The use of globally coupled weights in this proof does not change the algorithm. Proposition S4 is a deterministic inequality in the outcome path and the marginal laws D_{c_t} . The online forecaster uses independent fresh draws with exactly those marginals.

S3 Adaptive Outcomes and Simultaneous Concentration

Let U_t be an independent random seed used to generate $P_t \sim D_{c_{t-1}}$. Let $\mathcal{F}_{t-1}$ contain all previous seeds, forecasts, and outcomes. A nonanticipating adversary selects y_t as an $\mathcal{F}_{t-1}$ -measurable variable, after which P_t is generated from the fresh U_t . The order in which the unrevealed y_t and P_t are physically computed is immaterial. The requirement is that y_t not depend on U_t or P_t .

For a fixed loss, define

$$\mu_t = \mathbb{E}[\ell(P_t, y_t) \mid \mathcal{F}_{t-1}] = \mathbb{E}_{P \sim D_{c_{t-1}}} \ell(P, y_t),$$

with D_0 the first-round point mass. For every realized outcome path, the fixed-sequence proof in the main paper and Proposition S4 give

$$\sum_{t=1}^T \mu_t - \min_p \sum_{t=1}^T \ell(p, y_t) \leq 6 \sum_i \sqrt{n_i}. \quad (\text{S9})$$

This proves the expected adaptive guarantee after taking expectations. It also gives concentration because $Z_t = \ell(P_t, y_t) - \mu_t$ is a martingale difference.

Theorem S5 (Finite families and covers). *Let $\mathcal{L}_0$ contain M proper losses in $[-1, 1]$. Under the preceding adaptive protocol, with probability at least $1 - \delta$,*

$$\max_{\ell \in \mathcal{L}_0} \widehat{\text{Reg}}_\ell(T) \leq 6 \sum_i \sqrt{n_i} + \sqrt{2T \log(M/\delta)}. \quad (\text{S10})$$

For a fixed oblivious path,

$$\mathbb{E} \max_{\ell \in \mathcal{L}_0} \widehat{\text{Reg}}_\ell(T) \leq 6 \sum_i \sqrt{n_i} + \sqrt{2T \log M}. \quad (\text{S11})$$

More generally, suppose a class $\mathcal{L}$ has an ε -cover of size $M(\varepsilon)$ in uniform norm over $\Delta_K \times [K]$. Then with probability at least $1 - \delta$,

$$\sup_{\ell \in \mathcal{L}} \widehat{\text{Reg}}_\ell(T) \leq 6 \sum_i \sqrt{n_i} + 2\varepsilon T + \sqrt{2T \log(M(\varepsilon)/\delta)}. \quad (\text{S12})$$

Proof. For each loss, Z_t has conditional mean zero and lies in an interval of length two. Conditional Hoeffding's lemma yields

$$\mathbb{E} \left[e^{\lambda \sum_t Z_t} \right] \leq e^{\lambda^2 T / 2}. \quad (\text{S13})$$

Together with (S9), Chernoff's method and a union bound prove (S10). For a fixed path, apply the log-sum-exp bound to (S13):

$$\mathbb{E} \max_{\ell \in \mathcal{L}_0} \sum_t Z_{t,\ell} \leq \frac{\log M}{\lambda} + \frac{\lambda T}{2}.$$

Optimizing at $\lambda = \sqrt{2 \log M / T}$ gives (S11), with the case $M = 1$ following directly from mean zero. Finally, replacing a loss and its comparator by a uniformly ε -close cover element changes realized regret by at most $2\varepsilon T$. Apply (S10) to the cover to obtain (S12). $\square$

S4 Smooth-Loss Transfer in Full Detail

Let $c \neq 0$, $N = \sum_i c_i$, $q = c/N$, and $P \sim D_c$. Dirichlet covariance is

$$\text{Cov}(P_i, P_j) = \begin{cases} q_i(1 - q_i)/(N + 1), & i = j, \\ -q_i q_j / (N + 1), & i \neq j. \end{cases} \quad (\text{S14})$$

Taking the trace gives

$$\mathbb{E} \|P - q\|_2^2 = \frac{1 - \|q\|_2^2}{N + 1}. \quad (\text{S15})$$

These formulas remain valid on a proper face after deleting zero coordinates.

At round $t \geq 2$, $N = t - 1$ and empirical FTL predicts $q = c_{t-1}/(t - 1)$. For any β -smooth loss,

$$\begin{aligned} \mathbb{E}[\ell(P_t, y_t) - \ell(q, y_t)] &\leq \langle \nabla \ell(q, y_t), \mathbb{E}[P_t - q] \rangle + \frac{\beta}{2} \mathbb{E}\|P_t - q\|_2^2 \\ &\leq \frac{\beta}{2t}. \end{aligned}$$

Summing proves the transfer theorem without using properness. Properness is needed only to invoke the empirical FTL regret bound and to share the same best fixed comparator. More generally, the transfer applies to any smooth loss sequence for which empirical frequencies have a useful comparator guarantee.

S5 Class Aggregation and the All-Regime Lower Bound

We record the reduction used in Corollary 3 of the main paper. It makes explicit why the worst-loss rate is capped at T when the nominal alphabet is larger than the horizon.

Lemma S6 (Lifting through outcome aggregation). *Let $2 \leq r \leq K$, let $\pi : [K] \rightarrow [r]$ be surjective, and define $A_\pi : \Delta_K \rightarrow \Delta_r$ by*

$$(A_\pi p)_j = \sum_{i: \pi(i)=j} p_i.$$

If $\tilde{\ell} : \Delta_r \times [r] \rightarrow [-1, 1]$ is proper, then

$$\ell(p, i) = \tilde{\ell}(A_\pi p, \pi(i)) \tag{S16}$$

is proper on $\Delta_K \times [K]$. Moreover, every r -outcome regret lower bound transfers unchanged to K outcomes.

Proof. For $q \in \Delta_K$, the expected lifted loss is

$$\sum_{i=1}^K q_i \ell(p, i) = \sum_{j=1}^r (A_\pi q)_j \tilde{\ell}(A_\pi p, j).$$

Properness of $\tilde{\ell}$ makes this expression minimal when $A_\pi p = A_\pi q$. In particular, $p = q$ is a minimizer, so ℓ is proper.

Choose one representative $s_j \in \pi^{-1}(j)$ for each aggregated outcome. Given any K -outcome forecasting algorithm, map each forecast p_t to $A_\pi p_t$ and restrict the adversary to outcomes $s_1, \dots, s_r$. This produces a valid r -outcome forecasting algorithm. For every representative sequence $y_t = s_{z_t}$, (S16) gives equality of the incurred losses. Since A_π is onto,

$$\min_{p \in \Delta_K} \sum_t \ell(p, y_t) = \min_{u \in \Delta_r} \sum_t \tilde{\ell}(u, z_t).$$

Thus the regrets are identical, and any lower bound for the mapped algorithm transfers to the original one. $\square$

Corollary S7 (Dimension-capped lower bound). *For $K \geq 2$ and $T \geq 24$, every randomized K -outcome forecasting algorithm has an oblivious outcome path and a proper loss in $[-1, 1]$ for which*

$$\text{Reg}_\ell(T) = \Omega\left(\sqrt{\min\{K, T\}T}\right). \tag{S17}$$

Together with the main theorem, this proves the all-regime minimax rate.

Proof. If $K \leq T/12$, apply the lower bound of Luo, Senapati, and Sharan with $r = K$. Otherwise let $r = \lfloor T/12 \rfloor$. Then $2 \leq r \leq K$, $T \geq 12r$, and $r \geq \min\{K, T\}/24$. Apply their $\Omega(\sqrt{rT})$ lower bound and lift its proper loss and hard outcome path by Lemma S6. This gives (S17). $\square$

S6 Verification Audit

The accompanying script `verify_dirichlet.py` performs three independent checks.

First, it numerically integrates the exact total variation between $\text{Beta}(a, b)$ and $\text{Beta}(a + 1, b)$ and checks the variance upper bound in Lemma S1. Second, it samples representative Dirichlet laws and checks their mean and total squared variance against (S15). Third, it uses dynamic programming over all binary outcome paths to find exact worst expected regret for the Bayesian-bootstrap forecaster under squared loss and under the V-shaped proper loss used to show that empirical FTL can have linear regret.

The dynamic program keeps, for each count pair (a, b) , the largest cumulative expected loss among paths reaching that pair. The one-step expected loss depends only on (a, b) and the next outcome. At the horizon, the program subtracts the proper-loss value of the empirical-frequency comparator and maximizes over terminal count pairs. Table 1 reports representative values. The V-shaped column follows the expected square-root law, while squared loss follows the logarithmic law. These calculations are error checks only.

Table 1: Exact binary worst-path dynamic program.

T	V-shaped regret	regret/ $\sqrt{T}$	squared regret	regret/ $\log T$
16	2.798889	0.699722	1.771800	0.639042
64	5.951375	0.743922	2.464643	0.592621
256	12.272007	0.767000	3.157771	0.569463
1024	24.964856	0.780152	3.850917	0.555570